

Complete Agile Key Concepts Guide

This guide covers essential Agile and Scrum concepts for Scrum Masters, Project Managers, Delivery Managers, Agile Coaches, and Agile team members.

What is Agile?

Agile is an iterative and incremental approach to software delivery focused on collaboration, customer feedback, adaptability, and continuous improvement.

Agile Manifesto

The Agile Manifesto values individuals and interactions, working software, customer collaboration, and responding to change.

Scrum Framework

Scrum is a lightweight Agile framework consisting of roles, events, and artifacts used to deliver value iteratively.

Scrum Roles

Key Scrum roles are Product Owner, Scrum Master, and Development Team.

Product Owner

Responsible for maximizing product value, prioritizing backlog items, and stakeholder communication.

Scrum Master

Facilitates Scrum ceremonies, removes impediments, coaches the team, and ensures Scrum adoption.

Development Team

Cross-functional team responsible for delivering working increments.

Product Backlog

An ordered list of features, enhancements, bugs, and technical tasks.

Sprint

A time-boxed iteration usually lasting 1–4 weeks where the team delivers working software.

Sprint Planning

Ceremony where the team defines sprint goals and selects backlog items.

Daily Scrum

15-minute daily synchronization meeting for progress tracking and blocker identification.

Sprint Review

Meeting conducted to demonstrate completed work to stakeholders.

Sprint Retrospective

Team discusses improvements, lessons learned, and action items.

Definition of Done

Shared understanding of completion criteria for backlog items.

Definition of Ready

Checklist ensuring stories are prepared before entering a sprint.

User Stories

Simple requirement format: As a user, I want X so that Y.

Story Points

Relative estimation technique used to estimate effort or complexity.

Velocity

Average amount of work completed by a team during a sprint.

Burndown Chart

Visual representation of remaining work versus time.

Burnup Chart

Tracks completed work against total scope.

Kanban

Flow-based Agile methodology focusing on continuous delivery and limiting work in progress.

SAFe

Scaled Agile Framework used for implementing Agile at enterprise level.

PI Planning

Program Increment planning event in SAFe for alignment across teams.

RAID Log

Tracks Risks, Assumptions, Issues, and Dependencies.

Risk Management

Process of identifying, analyzing, and mitigating project risks.

Dependency Management

Managing tasks or teams dependent on each other.

Agile Metrics

Includes velocity, lead time, cycle time, defect leakage, and predictability.

Lead Time vs Cycle Time

Lead time measures request-to-delivery duration; cycle time measures active work duration.

Continuous Improvement

Regular refinement of processes to improve delivery quality and efficiency.

Servant Leadership

Leadership style focused on empowering and supporting the team.

Stakeholder Management

Building strong communication and alignment with stakeholders.

Jira

Popular Agile project management tool used for backlog tracking and sprint management.

Agile Estimation Techniques

Includes Planning Poker, T-shirt sizing, affinity estimation, and bucket system.

Impediments

Blockers preventing team progress.

Cross-functional Team

Team with all necessary skills to deliver end-to-end solutions.

Increment

Potentially shippable product output delivered after each sprint.

Technical Debt

Future rework caused by choosing short-term solutions.

DevOps

Collaboration between development and operations for faster and reliable delivery.

CI/CD

Continuous Integration and Continuous Delivery automate build and deployment pipelines.

Important Agile Metrics

Metric	Purpose
Velocity	Measure team throughput
Burndown	Track sprint progress
Lead Time	Request to delivery duration
Cycle Time	Actual development duration
Defect Leakage	Production defect analysis
Predictability	Planned vs completed work

Prepared for Agile learning, interview preparation, GitHub portfolio building, and professional knowledge sharing.